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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ADAPTIVE SOFTWARE ENGINEERING - (24CI3201)
A.Y. 2026 - 2027

TITLE OF THE PROJECT	**Cognix: Personal knowledge operating system with AI memory**	
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Abstract

Cognix is an AI-powered application designed to deliver intelligent, context-aware assistance by integrating modern full-stack development with advanced AI frameworks. The backend is developed using **Django (Python)**, while the frontend is built with **HTML, CSS, JavaScript, and Next.js (React.js)** to provide a fast, responsive, and user-friendly interface. **PostgreSQL** is used for structured data storage, and **Qdrant Vector Database** enables semantic search and efficient retrieval of relevant information through vector embeddings. The application leverages **LangChain** to build AI-driven workflows and connect large language models with external data sources, while **LangGraph** is used to orchestrate multi-step AI agent workflows and manage complex reasoning processes. By combining relational and vector databases with AI orchestration frameworks, Cognix can understand user queries, retrieve contextually relevant information, and generate accurate, meaningful responses. The project demonstrates the integration of modern technologies, AI frameworks, and vector search to build a scalable, intelligent, and efficient AI-powered platform

Signature of the Guide HOD